Problem Solving Strategies

Problem Statement:

A company wants to develop a **Customer Relationship Management (CRM) System** that manages customer data, tracks sales, and generates reports. The system should be designed using modular programming principles, where different functionalities are separated into distinct modules.

Functional Modules:

1. **Customer Management Module**
   - Stores customer details (name, email, phone).
   - Allows adding, updating, and deleting customer records.
2. **Sales Tracking Module**
   - Records sales transactions (customer, product, amount).
   - Tracks revenue and sales trends over time.
3. **Reporting Module**
   - Generates reports based on customer interactions and sales data.
   - Provides insights into customer behavior and sales performance.

In [1]:
```python
class CustomerManagement:
    def __init__(self):
        self.customers = []

    def add_customer(self, name, email, phone):
        customer = {"Name": name, "Email": email, "Phone": phone}
        self.customers.append(customer)
        return f"Customer {name} added successfully."

    def get_customers(self):
        return self.customers
```

In [2]:
```
class SalesTracking:
    def __init__(self):
        self.sales = []

    def record_sale(self, customer_name, product, amount):
        sale = {"Customer": customer_name, "Product": product, "Amount": amount}
        self.sales.append(sale)
        return f"Sale recorded for {customer_name}: {product} - ${amount}"

    def get_sales(self):
        return self.sales
```

In [3]:
```
class Reporting:
    def __init__(self, customer_mgmt, sales_mgmt):
        self.customer_mgmt = customer_mgmt
        self.sales_mgmt = sales_mgmt

    def generate_report(self):
        report = {
            "Total Customers": len(self.customer_mgmt.get_customers()),
            "Total Sales": len(self.sales_mgmt.get_sales()),
            "Revenue": sum(sale["Amount"] for sale in self.sales_mgmt.get_sales()),
        }
        return report
```

In [4]:
```
# Example Usage
if __name__ == "__main__":
    customer_mgmt = CustomerManagement()
    sales_mgmt = SalesTracking()
    report_mgmt = Reporting(customer_mgmt, sales_mgmt)

    # Adding Customers
    print(customer_mgmt.add_customer("Alice Johnson", "[email protected]", "123-456
    print(customer_mgmt.add_customer("Bob Smith", "[email protected]", "987-654-321

    # Recording Sales
    print(sales_mgmt.record_sale("Alice Johnson", "Laptop", 1200))
    print(sales_mgmt.record_sale("Bob Smith", "Smartphone", 800))

    # Generating Report
    print(report_mgmt.generate_report())
```

```
Customer Alice Johnson added successfully.
Customer Bob Smith added successfully.
Sale recorded for Alice Johnson: Laptop - $1200
Sale recorded for Bob Smith: Smartphone - $800
{'Total Customers': 2, 'Total Sales': 2, 'Revenue': 2000}
```